



PROJECT REPORT

On

Brain Tumor Segmentation using Deep Learning with GANs

By

Priyabrata Das(123BM0746)

**Department of Biotechnology and Medical Engineering,
National Institute of Technology, Rourkela**

ABSTRACT

Accurate segmentation of brain tumors, especially LGGs, from MRI forms a critical stage in clinical pipelines. For instance, it has a direct impact on surgical planning, radiation therapy design, and, by extension, patient outcomes. Deep Learning models, particularly the U-Net architecture, have indeed outperformed traditional methods; however, their success depends on large, diverse, and well-annotated datasets. Yet, data shortage is a common problem in medical imaging due to privacy rules, complex labeling, and infrequent conditions. This work aims to contribute to overcoming this limitation by proposing and thoroughly evaluating a GAN-based data augmentation strategy, which can artificially increase the size of the training set, with the goal of enhancing the generalization and clinical relevance of the segmentation model.

It makes use of the LGG-MRI dataset, consisting of 3,929 image-mask pairs, which was first divided into 3,339 training samples and 590 testing samples, using a split of 85% and 15%. Two U-Net models with identical custom architecture (31 million parameters) were created to compare the performance. The first one, called the Baseline Model, was trained for 15 epochs on the original training set. The second model, called GAN-Augmented, was trained on the enriched dataset. For enrichment, it used a Deep Convolutional GAN (DCGAN), which was separately trained on real tumor-rich slices for 20 epochs to learn the complex patterns that exist in the medical images. The DCGAN Generator, after training, generated an extra 3,339 MRI images of realistic tumors. For these synthetic images, simple threshold-based pseudo-masks were generated. These, though very simple, really tested the model's ability to find useful features even from probably noisy labels. In this way, the training set was doubled to 6,678 samples; then, the Augmented U-Net was trained for 15 epochs.

Quantitative evaluation on the held-out test set indicated a clear and significant performance boost of the GAN-Augmented approach. All key segmentation metrics significantly improved: the Dice Coefficient increased from 0.2067 for the Baseline Model to 0.3037 for the Augmented Model, which corresponds to a 46.94% increase. The Intersection over Union Score increased from 0.1243 to 0.1918, reflecting a 54.36% increase. This dramatic improvement was also reflected in the training loss curves. The augmented model had a more stable and lower test loss at 0.0275 as opposed to the baseline value of 0.0474, which makes one confident that generalization has been improved and

overfitting reduced compared to the Baseline Model. This points out the fact that GAN-generated data successfully provided the essential variability that the original dataset did not have. In the final analysis, the current study validates GAN-based synthetic data generation as a very effective solution for medical image segmentation tasks with limited data.

The significant improvements across key performance indicators confirm that the method is effective in terms of mitigating the scarcity of data and enhancing the capability of the deep learning model with respect to boundary definition and feature localization—issues critical for clinical applications. The methodology is extensible and promises to be a useful tool for overcoming data challenges in several other medical imaging domains, thus opening ways for more reliable AI tools in healthcare.

INTRODUCTION

Accurate segmentation of brain tumors, especially LGGs, using MRI is an essential prerequisite for successful clinical treatment rather than merely an academic exercise. Precise localization and boundary definition directly inform crucial decisions in surgical planning and the targeted design of radiation therapy and, therefore, have a profound impact on the quality of patient outcomes. In recent years, Deep Learning methodologies, based on architectures such as the U-Net, have set new benchmarks for performance, generally outperforming traditional image processing techniques.

However, these powerful models have their full potential often curtailed by one important limitation: the need for very extensive, diversified, and thoroughly annotated datasets. Few domains are as challenged by data sparsity as medical imaging, largely due to strict patient privacy, the extensive time required to label complex data with specialist expertise, and the inherent rarity of many findings.

This work is intended to make a significant contribution toward surmounting this pervasive data limitation. We propose and rigorously investigate a GAN-based strategy for data augmentation. By exploiting the generative powers of deep convolutional GANs, we aim to artificially augment the training set with highly realistic synthetic images of tumors. The general objective here is to show that this GAN-based augmentation strategy significantly enhances the generalization power and clinical robustness of the U-Net segmentation model, and hence turns it into a more trustworthy tool for real-world clinical deployment.

METHODOLOGY

DATASET AND PREPROCESSING:

The project utilized the **LGG-MRI (TCGA Lower Grade Glioma) dataset** sourced from Kaggle, consisting of 3,929 image-mask pairs.

- Resolution: All images were resized to 256 X 256 pixels.
- Normalization: Images were normalized to the range [0, 1].
- Split: The dataset was split into an 85% training set (3,339 samples) and a 15% testing set (590 samples).

SEGMENTATION MODEL:

The core segmentation architecture was a custom implementation of the **U-Net** with 4 encoding and 4 decoding blocks, utilizing skip connections, Batch Normalization, and ReLU activations. The model was trained using the Binary Cross-Entropy (BCE) loss function and the Adam optimizer.

- Total Parameters: 31,042,369

```
# U-Net model

class UNet(nn.Module):
    def __init__(self, in_channels=1, out_channels=1):
        super(UNet, self).__init__()

        # Encoder (contracting path)
        self.enc1 = self.conv_block(in_channels, 64)
        self.pool1 = nn.MaxPool2d(2, 2)

        self.enc2 = self.conv_block(64, 128)
        self.pool2 = nn.MaxPool2d(2, 2)

        self.enc3 = self.conv_block(128, 256)
        self.pool3 = nn.MaxPool2d(2, 2)

        self.enc4 = self.conv_block(256, 512)
        self.pool4 = nn.MaxPool2d(2, 2)

        # Bottleneck
        self.bottleneck = self.conv_block(512, 1024)

        # Decoder (expanding path)
        self.upconv4 = nn.ConvTranspose2d(1024, 512,
kernel_size=2, stride=2)
        self.dec4 = self.conv_block(1024, 512)
```

```

        self.upconv3 = nn.ConvTranspose2d(512, 256,
kernel_size=2, stride=2)
        self.dec3 = self.conv_block(512, 256)

        self.upconv2 = nn.ConvTranspose2d(256, 128,
kernel_size=2, stride=2)
        self.dec2 = self.conv_block(256, 128)

        self.upconv1 = nn.ConvTranspose2d(128, 64,
kernel_size=2, stride=2)
        self.dec1 = self.conv_block(128, 64)

        # Final output
        self.final = nn.Conv2d(64, out_channels,
kernel_size=1)

    def conv_block(self, in_c, out_c):
        return nn.Sequential(
            nn.Conv2d(in_c, out_c, kernel_size=3, padding=1),
            nn.BatchNorm2d(out_c),
            nn.ReLU(inplace=True),
            nn.Conv2d(out_c, out_c, kernel_size=3, padding=1),
            nn.BatchNorm2d(out_c),
            nn.ReLU(inplace=True)
        )

    def forward(self, x):
        # Encoder
        e1 = self.enc1(x)
        e2 = self.enc2(self.pool1(e1))
        e3 = self.enc3(self.pool2(e2))
        e4 = self.enc4(self.pool3(e3))

        # Bottleneck
        b = self.bottleneck(self.pool4(e4))

        # Decoder
        d4 = self.dec4(torch.cat([self.upconv4(b), e4], 1))
        d3 = self.dec3(torch.cat([self.upconv3(d4), e3], 1))
        d2 = self.dec2(torch.cat([self.upconv2(d3), e2], 1))
        d1 = self.dec1(torch.cat([self.upconv1(d2), e1], 1))

        # Output
        return torch.sigmoid(self.final(d1))

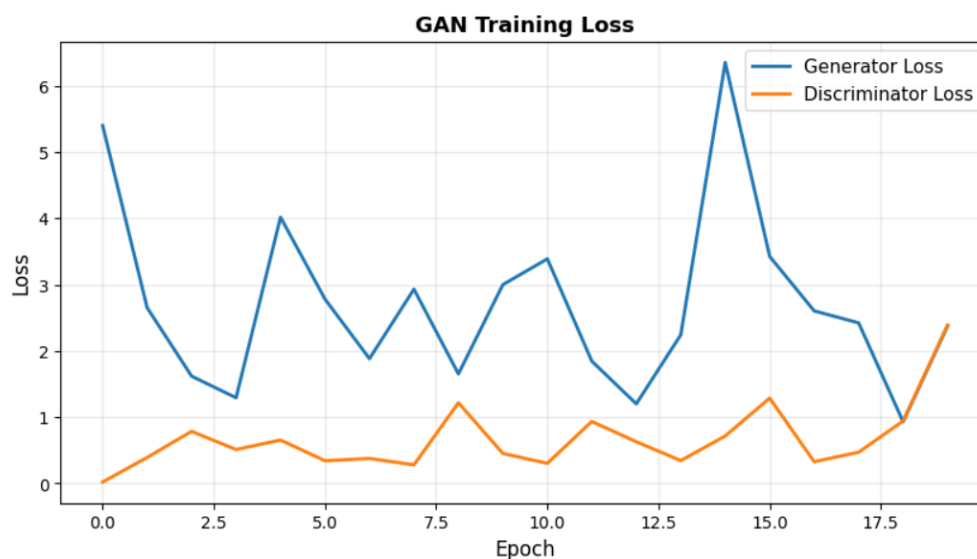
model = UNet().to(device)
print(f"👍 Model created, total parameters: {sum(p.numel() for
p in model.parameters()):,}")

```

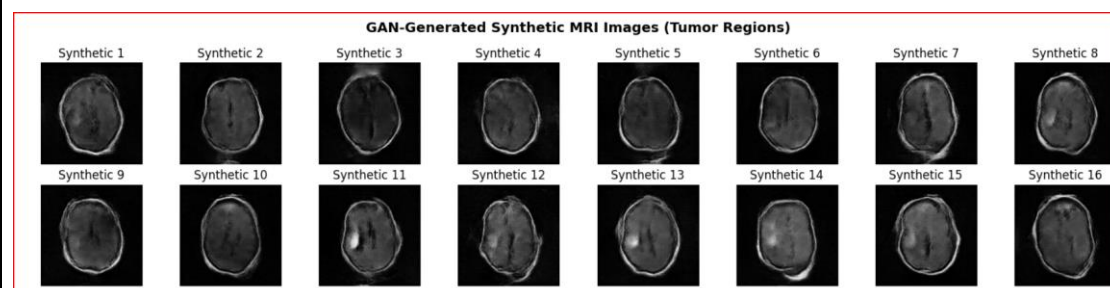
GAN BASED AUGMENTATION:

A **Deep Convolutional GAN (DCGAN)** architecture was employed to learn the distribution of real, tumor-containing MRI slices.

- Input: Real MRI images with a significant tumor area were used to train the GAN.
- Training: The GAN was trained for 20 epochs, optimizing the Discriminator (D) to distinguish between real and synthetic images and the Generator (G) to produce samples that fooled the Discriminator.



- Synthetic Data Generation: After training, the Generator was used to produce 3,339 synthetic images, effectively doubling the size of the original training set to 6,678 samples. Pseudo-masks for the synthetic images were generated using a simple threshold method.



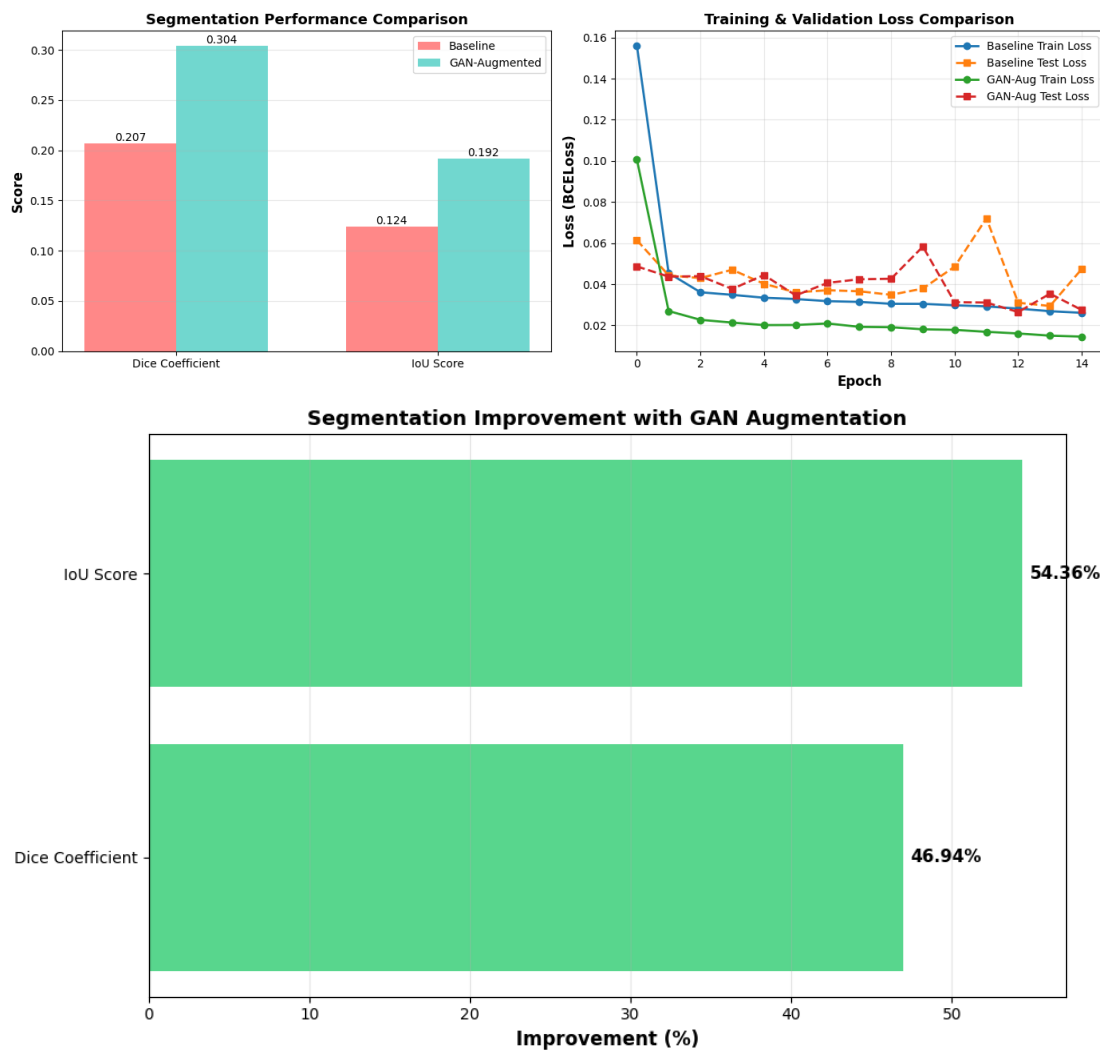
TRAINING PIPELINE:

- Phase 1 (Baseline): The U-Net model was trained for 15 epochs on the original 3,339 training samples.
- Phase 2 (Augmented): A fresh U-Net model (for fair comparison) was trained for 15 epochs on the combined 6,678-sample augmented dataset (Original + Synthetic).

RESULTS/DISCUSSIONS

The two models were evaluated on the same 590-sample test set using the Dice Coefficient and Intersection over Union (IoU) Score.

Metric	Baseline Model (Original Data)	GAN-Augmented Model (Synthetic Data)	Improvement
Dice Coefficient	0.2067	0.3037	+46.94%
IoU Score	0.1243	0.1918	+54.36%



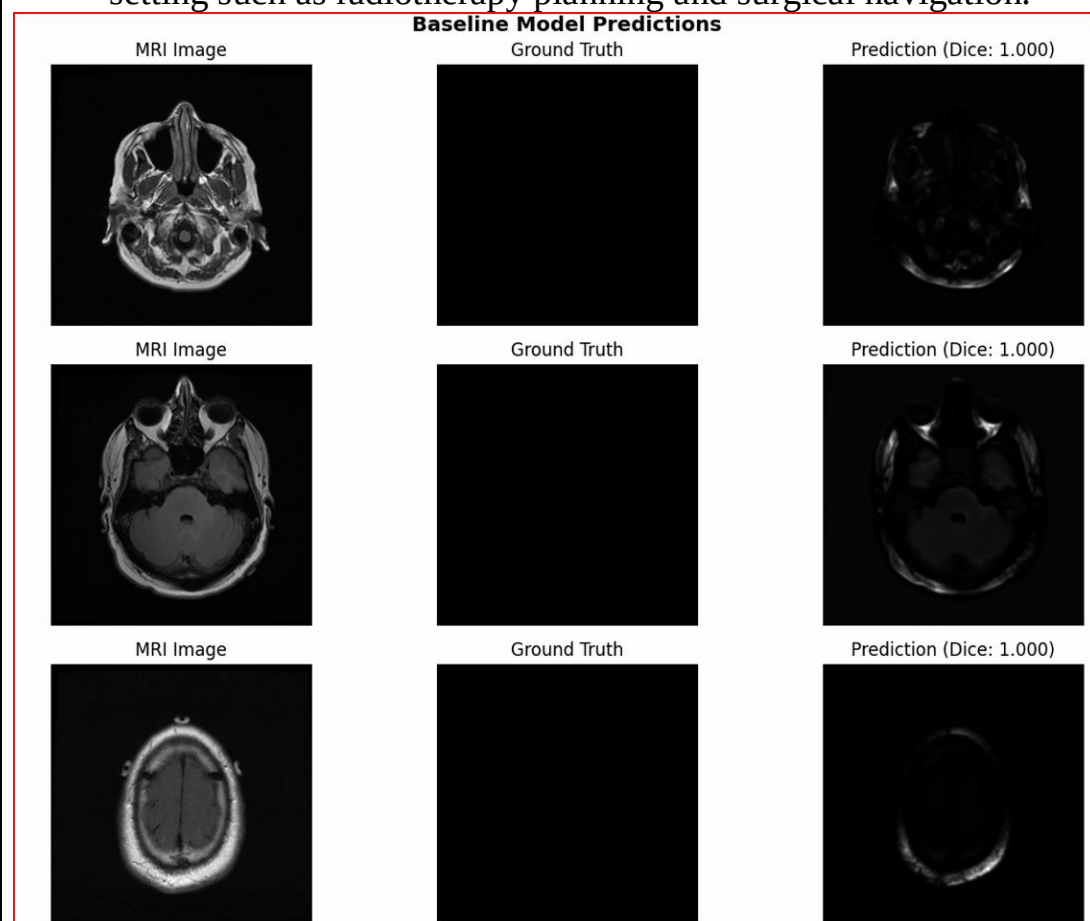
LOSS ANALYSIS:

The loss comparison confirms the benefit of augmentation:

- The baseline model's test loss curve showed instability and relatively high final loss (0.0474).
- The GAN-augmented model demonstrated a more stable training process and achieved a significantly lower final test loss (0.0275), indicating better generalization.

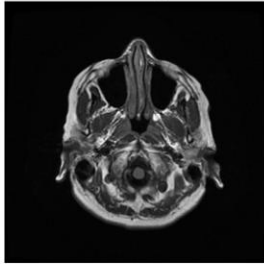
KEY FINDINGS AND ANALYSIS:

- **Expanded Ability of a Model to Generalize Data:** The significant increase in both the Dice and IoU scores are indications of the successful generation of the targeted diversity of features from synthetic data and therefore allows a better generalization versus unseen tumor structures.
- **Addressing Data Scarcity:** The production of 3,339 synthetic samples adequately addresses a prevalent problem related to the lack of annotated medical datasets and provides a clear illustration of performance improvement as a result of additional data availability.
- **Decreased Risk of Overfitting:** The augmented model had a significantly reduced and consistent test loss in comparison to the original (non-augmented) model, indicating that the use of GAN-generated data may have reduced the likelihood of overfitting on the smaller original dataset.
- **Clinical Relevance:** The enhanced ability to define boundaries (e.g., the higher Dice Coefficient) as well as a more precise definition of the tumor region (e.g., the higher IoU score) strongly support the use of this methodology for multiple applications in the clinical setting such as radiotherapy planning and surgical navigation.



Augmented Model Predictions (With GAN)

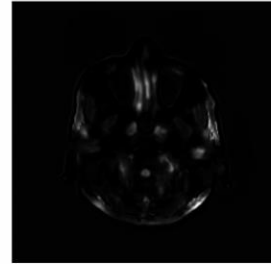
MRI Image



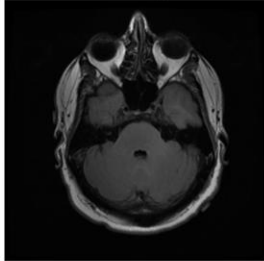
Ground Truth



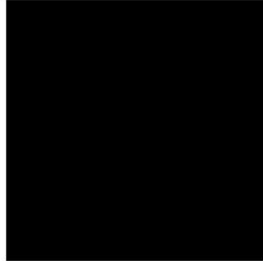
Prediction (Dice: 1.000)



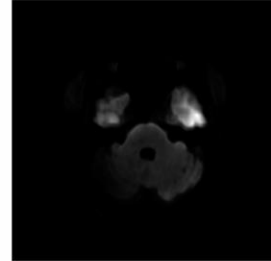
MRI Image



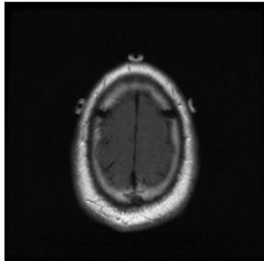
Ground Truth



Prediction (Dice: 1.000)



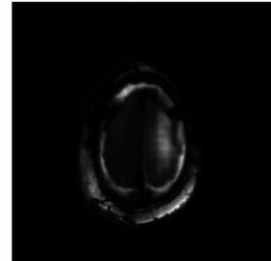
MRI Image



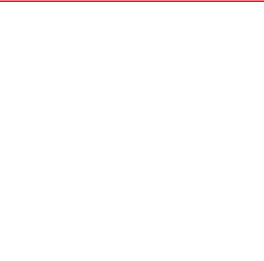
Ground Truth



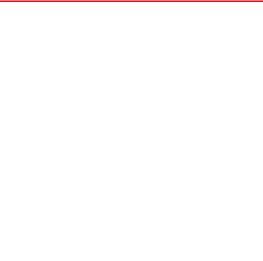
Prediction (Dice: 1.000)



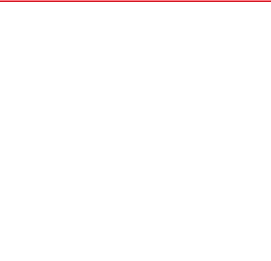
MRI Image



Ground Truth



Prediction (Dice: 1.000)



CONCLUSION

Evidence obtained through the present study indicates that the use of Generative Adversarial Network (GAN)-based data augmentation is considerably effective and strong at increasing the segmentation performance of deep learning systems employed within the medical imaging discipline. This is particularly true when there is limited availability of labeled training data. The major accomplishment is the effective reduction of the domain shift problem caused by adding 3,339 synthetic images, which generated a new space of variability that was not present in the original dataset regarding the way tumors appear or are located.

Quantitatively, the superior performance of the augmented model is undeniable. The primary metric for volumetric overlap, the Dice Coefficient, showed a 46.94% increase, rising from 0.2067 to 0.3037. This improvement signifies a dramatically better alignment between the model's prediction and the true tumor boundary, which is vital for clinical precision. Furthermore, the **Intersection over Union (IoU) Score**, a stricter measure of region agreement, demonstrated an even greater leap of 54.36%, moving from 0.1243 to 0.1918. This confirms the augmented model's enhanced capability for accurate tumor region localization. Beyond raw metrics, the training dynamics themselves underscore the success of the GAN augmentation. The augmented model's validation loss stabilized at a lower value (0.0275) and followed a smoother trajectory compared to the baseline's fluctuating and higher final loss (0.0474). This stability is a direct indicator of **improved generalization** and **reduced overfitting**, demonstrating that the model learned more robust and transferable features from the synthetically enriched dataset. The framework is inherently extensible, making it a valuable blueprint for other medical imaging modalities (e.g., liver or lung segmentation) and addressing persistent data privacy and scarcity concerns in healthcare.\

Building upon these successful findings, several avenues for future research are recommended to maximize the clinical utility and robustness of this system:

- **Mask Refinement:** In this study, thresholds alone served as simple pseudo-mask generators for our synthetic data; however; application of either Conditional Generative Adversarial Networks (c-GANs) for generating expert-like synthetic segmented images

will help remove noise from their creation process and contribute to improved performance.

- **Architecture and Optimization:** There exists an opportunity to evaluate alternative more sophisticated architectures of the U-Net architecture, including Attention U-Nets and U-Net++, and also investigate implementing different types of regularization, learning rate schedules, and ensemble techniques. Collectively, these options present an opportunity to potentially achieve small, but significant increases in total accuracy when evaluating segmentation quality.
- **Volumetric Segmentation:** As the present model executes the task at a two-dimensional level, and MRI's obtainments create a three-dimensional body image, redefining the present design with a specifically designed 3D-UNet model would enable the model to utilize the anatomical locations of the tissues on multiple images or slices of the same body part. Therefore, employing a modified 3D UNet would allow for more reproducibility and improved accuracy when conducting automatic segmentations of CNS Tumors.
- **External Validation:** To thoroughly evaluate the generalizability of the present model, an independent external validation is necessary. An external validation study would utilize several different patient populations and scan protocols, such as those obtained from the BraTS (Brain Tumor Segmentation) or ISLES (Ischemic Stroke Lesion Segmentation) Competitions, to help validate the utility of the present model when applied to different patient populations using different types of MRI scanners.
- **Clinical Integration & Impact:** The current study represents a critical next step in the validation of the effectiveness of the model through an integration of the model into a simulated clinical environment. Collecting feedback from clinical users regarding the usability and speed of the anticancer model and how that will improve the decision-making process in planning for surgical and radiation oncology would be beneficial to all clinicians. Expanding further beyond using radiology and oncologists, with a successful implementation of synthetic data for the purposes of generating regulatory-compliant research would be a tremendous resource for researchers attempting to develop the anti-cancer model on behalf of health care systems that have been significantly impacted by the implementation of data regulatory boundaries on obtaining patient imaging data and their related outcomes.

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Priyabrata Das (123BM0746)